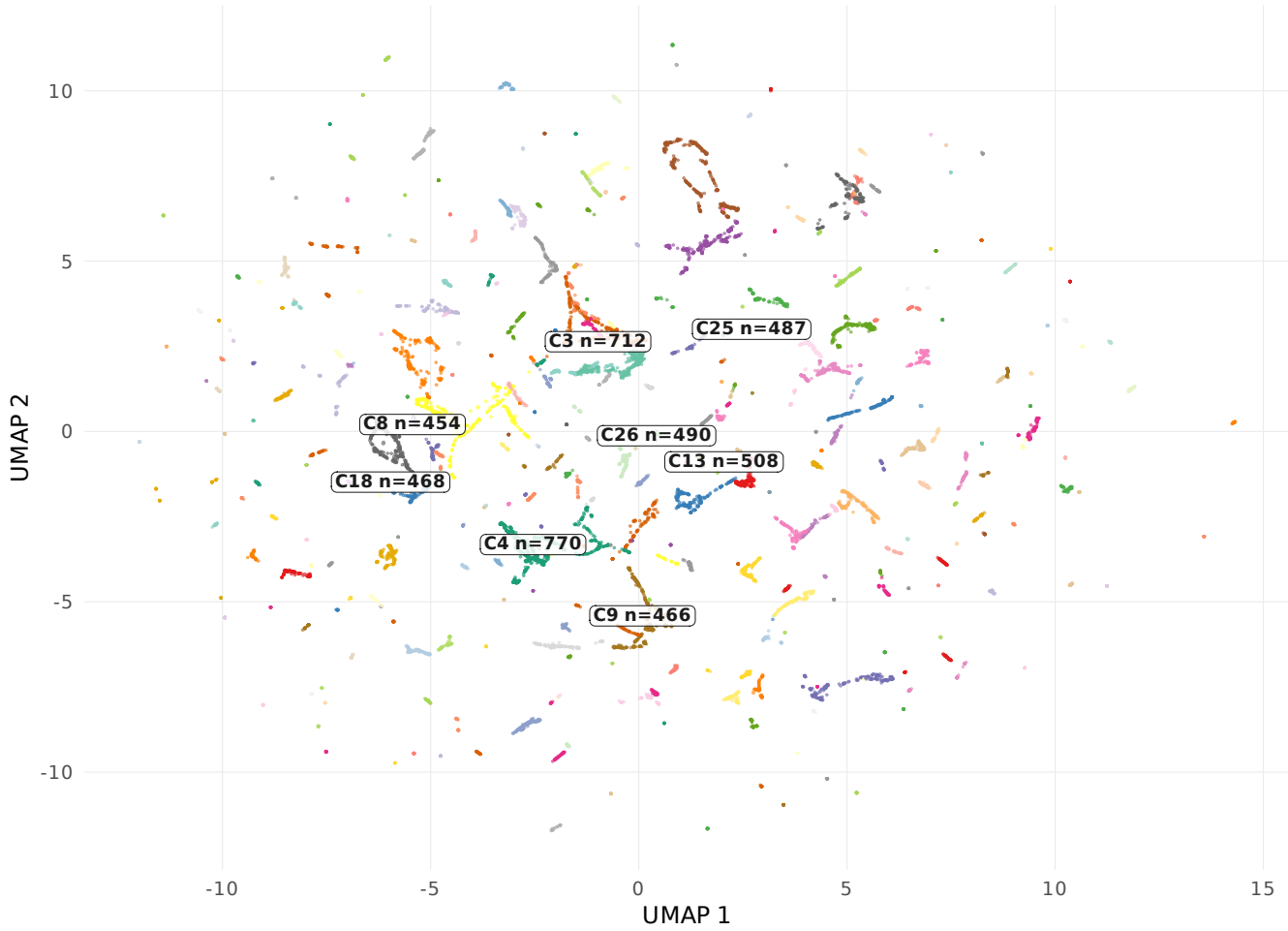


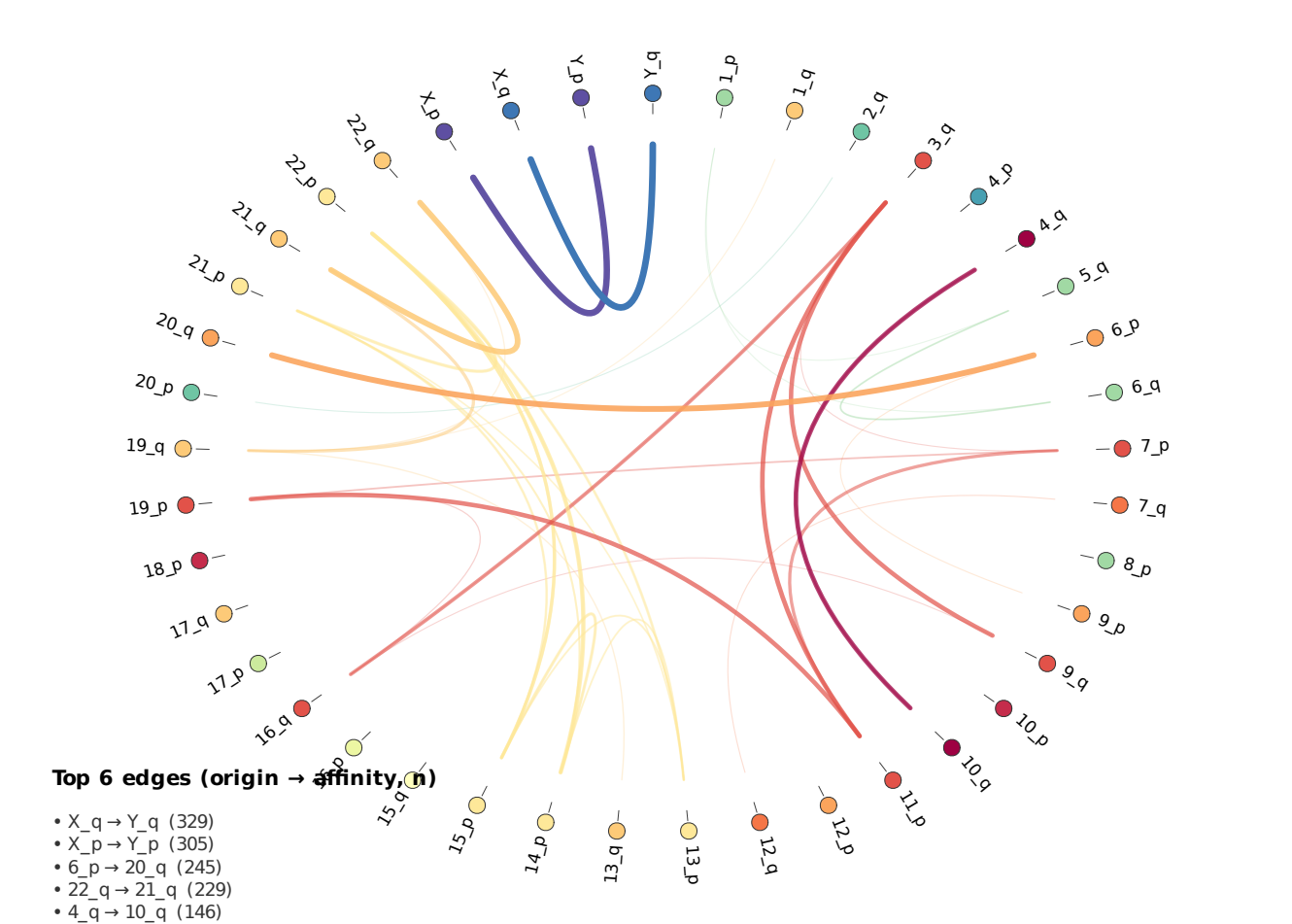
ED2a — UMAP of 15,668 PHRs by 50-community partition

Leiden k = 75, resolution = 0.8, modularity 0.97; top 8 communities labelled



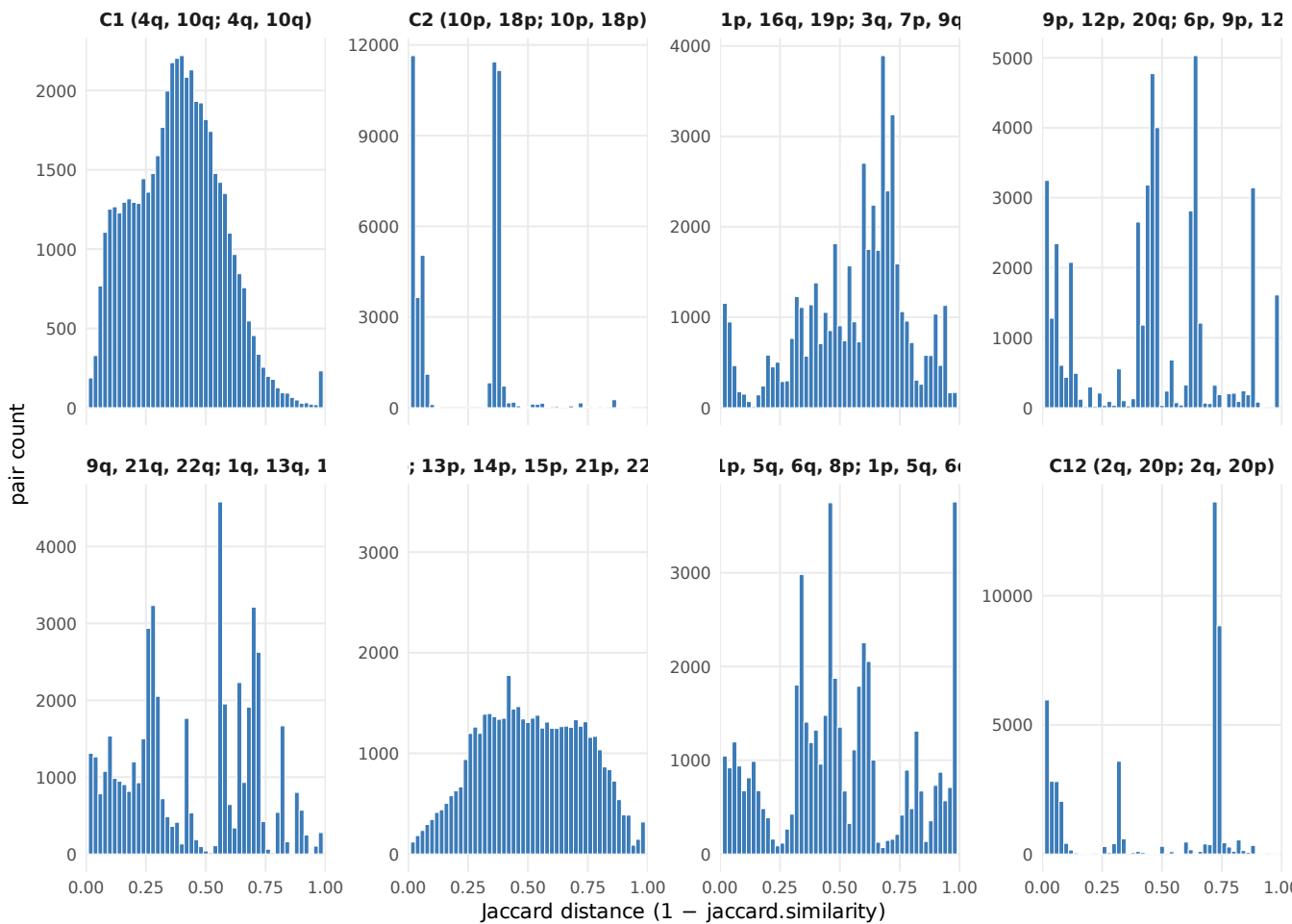
ED2c — cross-arm affinity (sequences absorbed into a foreign arm community)

41 arms (origin → affinity); chord width ∝ √(#sequences); colour = arm-Leiden community



ED2b — within-community Jaccard distance distributions

8 arm-level communities (≤50,000 random pairs each); bimodality marks allele/paralog separation



ED2d — Arm-Leiden (15) vs Sequence-Leiden (50) confusion (ARI 0.35, NMI 0.76)

n = 15,668 PHRs; rows = arm-level communities, columns = sequence-level communities

